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A DATASETS AND CATEGORIES

Table 4 provides a summary of the 24 datasets and their respective sizes (number of questions) used in our research. These datasets evaluate the knowledge and reasoning abilities of LLMs on a wide range of subject domains.

Specifically, we selected 22 datasets from 15 different subcategories of the *MMLU* benchmark. Additionally, we incorporated the *MBPP* dataset, which is designed to assess the proficiency of LLMs in generating Python programs for specific coding problems such that they pass predefined unit tests successfully. Furthermore, we included the *Sports Understanding* dataset, which assesses multi-hop reasoning skills in the context of sports, actions, and athletes.

Due to resource constraints, we randomly sample 250 questions from the larger datasets, which include moral scenarios, professional medicine, professional law, professional accounting, and professional psychology. For all datasets, we make use of the official test partitions in our evaluations.

Dataset	Size
abstract algebra	99
anatomy	134
college biology	143
college chemistry	99
college computer science	99
college mathematics	99
college physics	101
computer security	99
conceptual physics	234
high school chemistry	202
high school government and politics	192
high school world history	236
human sexuality	130
logical fallacies	162
machine learning	111
management	102
mbpp	257
moral scenarios	250
professional accounting	250
professional law	250
professional medicine	250
professional psychology	250
sociology	200
sports understanding	250

Table 4: The 24 datasets with their sizes (number of questions) that comprise our evaluation suite.

We categorized the 24 datasets into 5 broad categories. Table 5 displays the sizes (number of questions) for each of these categories. The datasets associated with each category are presented in Table 6.

Category	Size
Computer Science	566
Formal Science	198
Natural Science	1293
Social Science	1642
Ethics	250

Table 5: The 5 broad categories with their sizes (number of questions) that we use in our study.